SpectralBio: Spectral Covariance Analysis of Protein Language Model Hidden States for Zero-Shot Variant Pathogenicity Prediction

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Abstract. Zero-shot missense scoring with protein language models is usually framed as a sequence-likelihood problem. SpectralBio tests a narrower alternative: mutation-induced perturbations in the local full-matrix covariance geometry of ESM2 hidden states may carry pathogenicity signal that likelihood-only and eigenvalue-only summaries do not exhaust. The manuscript centers a stronger-baseline BRCA2 audit, while the public executable replay center remains the TP53 canonical benchmark. On BRCA2, adding covariance-aware geometry to a five-model ESM-1v ensemble improves AUC from **0.6324** to **0.6890**, for paired gain **0.0566**, paired 95% bootstrap CI [**0.0131, 0.1063**], and empirical permutation $p = 0.0010$. On the frozen TP53 canonical benchmark, the released pair $0.55 \cdot$ **FrobDist** $+ 0.45 \cdot$ **LL Proper** reaches **AUC=0.7498**, and repeated nested cross-validation places the fixed released weight on a stable out-of-fold plateau (**0.7510** vs. **0.7485** for re-tuned alpha). Across a support-ranked top-25 feasible panel derived from a 15,752-gene ClinVar scan, 10 genes show positive pair-versus-likelihood lower bounds, 2 are clearly negative, and the remainder are ambiguous. SpectralBio is therefore a bounded representational result and reproducibility artifact: covariance-aware hidden-state geometry can improve a stronger baseline in a benchmark-qualified gene, survives executable audit on TP53, and behaves as a structured rather than universal phenomenon.

1 Introduction

Zero-shot missense prediction is usually reduced to sequence likelihood: a mutation is scored by how surprising the mutant residue appears under a protein language model (PLM) [2]. This framing has been highly useful, but it leaves open a narrower mechanistic question. A missense change may be only moderately surprising at the token level while still reorganizing the local geometry of hidden states. If that geometry carries benchmark-relevant pathogenicity signal, then scalar likelihood alone is an incomplete readout.

SpectralBio studies this question through local covariance analysis of ESM2 hidden states. For each mutation, we compare wild-type and mutant covariance matrices in a window centered on the altered residue and derive full-matrix spectral features that can be combined with likelihood-based scores. The repository deliberately separates the manuscript scientific center from the frozen executable replay center. The flagship scientific result is a stronger-baseline augmentation audit on BRCA2; the only frozen public replay surface remains TP53; the support-ranked top-25 feasible panel is the performance-blind breadth surface; and protocol sweep plus BRCA1 failure analysis define the current boundaries of the method.

The contribution is therefore bounded and multi-surface. On BRCA2, covariance-aware geometry improves a stronger external ESM-1v baseline. On TP53, the released pair is stable under nested validation and remains machine-checkable through a frozen public artifact. Across the support-ranked top-25 panel, outcomes are heterogeneous rather than uniformly favorable. Together these surfaces support a representational claim about covariance-aware hidden-state geometry, not a universal pathogenicity or clinical-deployment claim.

2 Related Work

Large PLMs such as ESM-2 have become strong zero-shot baselines for mutation-effect prediction [1, 2]. ProteinGym broadened this benchmarking culture by emphasizing explicit comparable evaluation surfaces rather than anecdotal gene-specific wins [5]. In that literature, the dominant readout remains sequence likelihood or closely related scalar summaries.

Spectral analysis of hidden-state geometry offers a complementary perspective. SpectralGuard showed that covariance-level perturbations in model representations can carry discriminative signal beyond surface probabilities [4]. SpectralBio transfers that intuition to zero-shot missense scoring and asks whether full-matrix covariance perturbations provide usable pathogenicity signal beyond both likelihood-only and eigenvalue-only summaries.

3 Method

3.1 Evidence Surfaces and Backbone

The public canonical backbone is ESM2-150M (`esm2_t30_150M_UR50D`; 30 layers, hidden size 640) in inference mode with no fine-tuning [1]. Missense variants are sourced from ClinVar after binary pathogenic/benign filtering [3]. The frozen executable replay surface is TP53 ($N = 255$; 115 pathogenic, 140 benign), with an auxiliary bounded transfer surface on a fixed BRCA1 subset ($N = 100$) without retraining. The manuscript-facing flagship surface is a BRCA2 stronger-baseline audit against a five-model ESM-1v ensemble, complemented by a support-ranked top-25 feasible panel derived from a 15,752-gene ClinVar scan and by protocol/boundary analyses.

3.2 Covariance Features and Score Construction

For a wild-type sequence $\mathbf{S}_{\mathrm{WT}}$ and mutant sequence $\mathbf{S}_{\mathrm{MUT}}$ at position p, we extract a ± 40 residue local window around the mutation and compute hidden states for each ESM2 layer. For layer $\ell \in \{1, \ldots, L\}$ with hidden matrix $\mathbf{H}^{(\ell)} \in \mathbb{R}^{w \times d}$, we form residue-level covariance $\mathbf{C}^{(\ell)} = \mathrm{Cov}(\mathbf{H}^{(\ell)})$. Three features are then averaged across layers:

$$\mathrm{FrobDist} = \frac{1}{L} \sum_{\ell=1}^{L} \left\| \mathbf{C}_{\mathrm{MUT}}^{(\ell)} - \mathbf{C}_{\mathrm{WT}}^{(\ell)} \right\|_F \tag{1}$$

$$\mathrm{TraceRatio} = \frac{1}{L} \sum_{\ell=1}^{L} \left| \frac{\mathrm{tr}(\mathbf{C}_{\mathrm{MUT}}^{(\ell)})}{\mathrm{tr}(\mathbf{C}_{\mathrm{WT}}^{(\ell)})} - 1 \right| \tag{2}$$

$$\text{SPS-log} = \frac{1}{L} \sum_{\ell=1}^{L} \left\| \log |\boldsymbol{\lambda}_{\mathrm{MUT}}^{(\ell)}| - \log |\boldsymbol{\lambda}_{\mathrm{WT}}^{(\ell)}| \right\|_2^2 \tag{3}$$

where $\boldsymbol{\lambda}^{(\ell)}$ denotes the covariance eigenvalues. FrobDist and TraceRatio retain matrix-level information; SPS-log keeps only eigenvalue summaries.

The likelihood branch uses the masked-language-model score

$$\begin{aligned} \text{LL Proper}(v) = {} & \log P_{\mathrm{ESM2}}(r_p^{\mathrm{wt}} \mid \mathbf{S}_{\mathrm{WT}}) \\ & - \log P_{\mathrm{ESM2}}(r_p^{\mathrm{mut}} \mid \mathbf{S}_{\mathrm{WT}}) \end{aligned} \tag{4}$$

Within a scored surface, features are MinMax-normalized and combined as $s = \alpha f_1 + (1 - \alpha) f_2$ over an alpha grid with step 0.05. The released TP53 pair is fixed at $0.55 \cdot \mathrm{FrobDist} + 0.45 \cdot \text{LL Proper}$.

3.3 Evaluation

We report AUC-ROC with 95% nonparametric bootstrap intervals ($B = 1000$, seed 42). Pair-versus-baseline deltas use paired bootstrap resampling of the same variant indices. TP53 and BRCA2 validation audits use repeated 5×5 stratified cross-validation, comparing the released fixed weight against re-tuned alpha on held-out folds. The BRCA2 stronger-baseline falsification test permutes covariance alignment against the ESM-1v branch while holding labels fixed, yielding an empirical tail probability for the observed gain.

4 Results

4.1 Flagship Stronger-Baseline Result on BRCA2

The strongest single result in the manuscript is the BRCA2 augmentation audit. Replacing the internal likelihood comparator with a five-model ESM-1v ensemble makes the question harder and cleaner: does covariance still add signal against a stronger external zero-shot baseline? On BRCA2, the answer is yes. The fixed covariance-plus- ESM-1v score improves AUC from 0.6324 to 0.6890, for paired gain 0.0566 with paired 95% CI [0.0131, 0.1063]. Under covariance permutation, that gain disappears ($p = 0.0010$), arguing that the improvement is not a generic metric artifact. In the separate benchmark-qualification audit, BRCA2 also reaches fixed nested mean AUC 0.7448 versus 0.6938 for likelihood-only, making it the only current non-anchor gene that satisfies the repository's benchmark-promotion rule.

4.2 TP53 Canonical Validation Anchor

TP53 remains the only frozen public canonical replay surface and the executable validation anchor of the paper. On that benchmark, the released pair $0.55 \cdot \mathrm{FrobDist} + 0.45 \cdot$ LL Proper reaches AUC 0.7498 with reproducibility delta 0.0. Repeated nested cross-validation places the fixed released weight on a stable out-of-fold plateau (mean AUC 0.7510) rather than below the re-tuned alternative (0.7485), which directly addresses the simplest same-surface overfitting criticism.

The TP53 surface also clarifies what part of covariance is useful. Matrix-level features outperform eigenvalue-only summaries: TraceRatio reaches 0.6242 and FrobDist 0.6209, while SPS-log reaches 0.5988 and other SPS variants are lower. The best TP53 pair also exceeds either branch alone, with FrobDist and LL Proper combining mechanistically distinct information: representational displacement and sequence-level surprisal.

4.3 Breadth and Boundary Surfaces

The manuscript no longer relies on a favorable companion-gene narrative. The performance-blind breadth surface begins from a 15,752-gene ClinVar scan, applies predeclared support thresholds, yields 446 threshold-passing genes, and realizes a support-ranked feasible top-25 panel with 10,992 scored variants. Within that panel, 10 genes show positive pair-versus-likelihood lower confidence bounds, 2 are clearly negative, and 13 are ambiguous. This is weaker than universal transfer, but stronger than a hand-built all-positive companion panel because heterogeneity is now explicit.

Boundary analyses make the hard cases interpretable instead of hidden. A 192-configuration sweep across checkpoints, window radii, layer protocols, and alpha handling shows that covariance utility is checkpoint-, window-, and layer-sensitive rather than a trivial 150M-only artifact. BRCA1 remains the most visible hard-negative surface, but its behavior is structured rather than uniformly contradictory. The released BRCA1 transfer AUC of 0.9174 is kept as bounded auxiliary evidence on a fixed subset without retraining; it is not promoted to a co-primary manuscript center.

Table 1: **Headline evidence surfaces in the final manuscript.** The manuscript center is BRCA2, the frozen executable center is TP53, and the remaining surfaces test breadth and boundaries rather than replacing the canonical replay contract.

Surface	Key Result	Interpretation
BRCA2 stronger-baseline audit	ESM-1v 0.6324 → covariance + ESM-1v 0.6890; paired gain 0.0566; CI [0.0131, 0.1063]; $p = 0.0010$	Flagship evidence that covariance-aware geometry can improve a stronger external baseline
TP53 canonical replay	Released pair AUC 0.7498; nested fixed 0.7510 vs. 0.7485 for re-tuned alpha	Executable validation anchor showing the released score is stable rather than a same-surface spike
Support-ranked top-25 panel	10 positive lower bounds, 2 clearly negative, 13 ambiguous across 10,992 variants	Performance-blind breadth surface showing structured heterogeneity rather than hand-picked favorable transfer
Protocol and BRCA1 boundary analyses	192 scored configurations; BRCA1 remains structured boundary case; fixed BRCA1 transfer AUC 0.9174 is auxiliary only	Boundary evidence defining where the method is sensitive, not a co-primary flagship benchmark

5 Discussion

Taken together, these results support a bounded representational claim. The strongest evidence is not that covariance beats every baseline everywhere, but that it survives a stronger-baseline test on BRCA2, remains executable and auditable on TP53, appears on a performance-blind breadth surface with mixed outcomes, and behaves under protocol perturbation as a structured phenomenon rather than a universal law. This is why the paper is framed as a research reproducibility artifact plus scientific audit surfaces, not as a universal pathogenicity predictor or clinical deployment recipe.

The role split across surfaces is important. BRCA2 is the manuscript's flagship scientific result and the clearest direct evidence that covariance-aware hidden-state geometry can add signal beyond a stronger external baseline. TP53 remains the only frozen public canonical replay surface and therefore the validation anchor that makes the claim challengeable. The top-25 panel shows that the paper is not built from favorable hand-selection, while the protocol sweep and BRCA1 analysis define concrete boundary conditions. BRCA2 is therefore best understood as the next canonicalization target under the stated promotion rule, not as an already-frozen replacement for TP53.

6 Reproducibility

The cold-start public replay path is `uv sync -frozen` followed by `uv run spectralbio canonical`, which materializes the frozen TP53 artifact from bundled inputs and bundled score references. Optional auxiliary validation then runs `uv run spectralbio transfer`, `uv run spectralbio verify`, and `python -m uv run python scripts/preflight.py`. The BRCA2 flagship result, the support-ranked top-25 panel, the protocol sweep, and the BRCA1 failure analysis are released as public scientific audit surfaces through the repository's paper-facing text and notebooks rather than through the default CPU-only canonical replay path.

7 Conclusion

SpectralBio shows that full-matrix covariance geometry of PLM hidden states can carry recoverable zero-shot pathogenicity signal beyond likelihood-only and eigenvalue-only summaries. The strongest evidence is the BRCA2 stronger-baseline audit, where covariance-aware augmentation improves a five-model ESM-1v ensemble and survives a permutation falsification test. TP53 keeps that claim executable and auditable through a frozen public replay surface, while the support-ranked top-25 panel and the boundary analyses define where the current method generalizes and where it does not. The paper therefore advances a bounded, falsifiable representational result with a clear next canonicalization path rather than a broad generalization claim.

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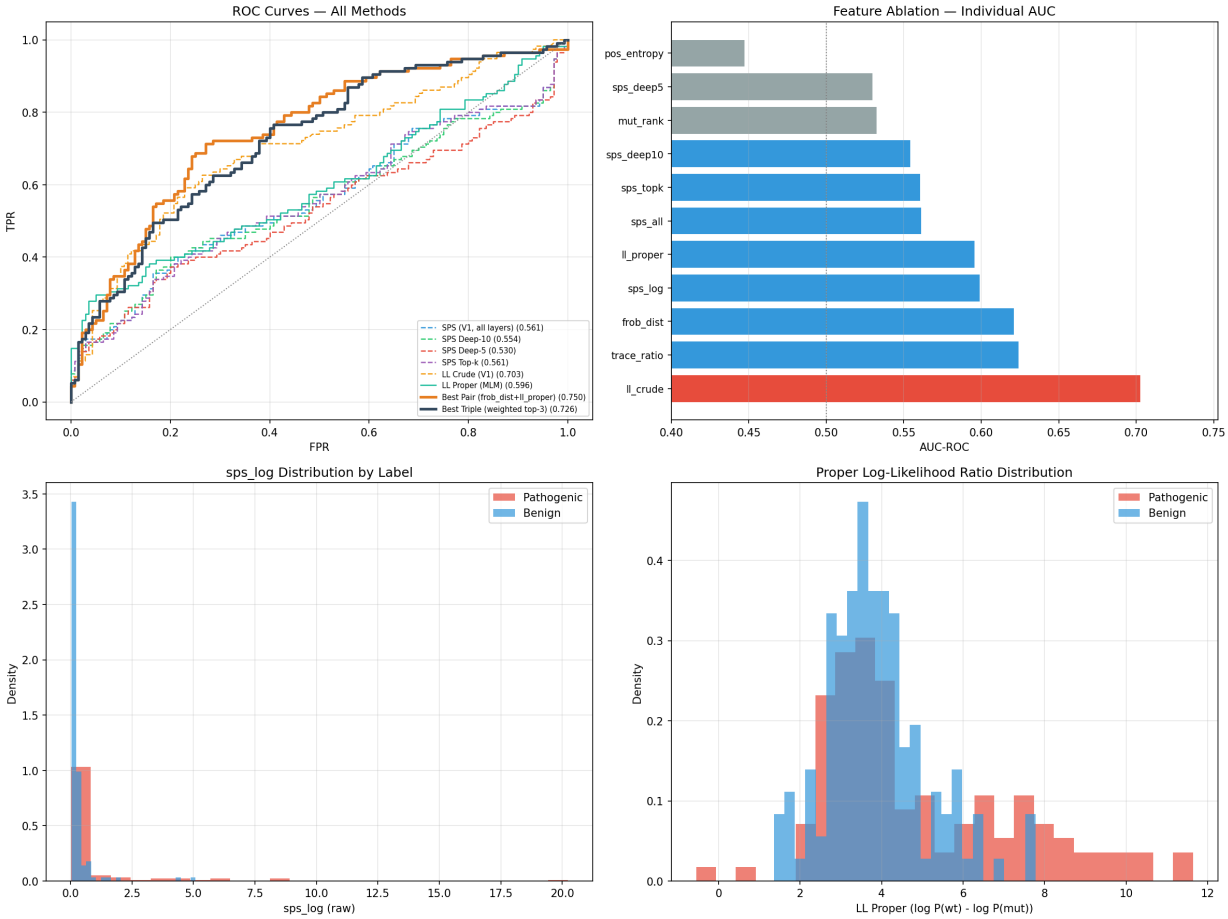

Figure 1: **TP53 canonical executable validation surface.** The released pair (orange) remains above the individual covariance and likelihood branches on the frozen TP53 replay path and materializes as a machine-checkable artifact under `outputs/canonical/`. This figure summarizes the executable anchor that keeps the manuscript's covariance claim challengeable on a public, deterministic surface.